AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 4 · WORKBOOK

Sampling and Variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected sampling record, with a small attention extension.

Trace boundaries, calculate probabilities, compare a draw and specify a fair experiment.

Keep one habit: Retain token order, exact intervals, draws, transformations, outcomes and separate reference checks.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Tokens and Context

Predictive and Generative AI

MAIN LESSON · SECTION 12

W1 · Trace interval boundaries

> Reject the draw1 under the stated half-open contract.

Your evidence record

- Main table order red,blue,green with probabilities0.6,0.3,0.1. Intervals[0,0.6),[0.6,0.9),[0.9,1). T=0.5 squares and normalises:18/23,9/46,1/46; boundaries18/23,45/46,1. Separate pictured A/B probabilities0.8/0.2 become16/17,1/17. Do not substitute those boundaries into the main table.

- At T=1 select tokens for u=0,0.6,0.9 and 0.999. Explain the handling of u=1.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Compute the adjusted table

> Square, total and normalise with exact fractions.

Your evidence record

- Main table order red,blue,green with probabilities0.6,0.3,0.1. Intervals[0,0.6),[0.6,0.9),[0.9,1). T=0.5 squares and normalises:18/23,9/46,1/46; boundaries18/23,45/46,1. Separate pictured A/B probabilities0.8/0.2 become16/17,1/17. Do not substitute those boundaries into the main table.

- At T=0.5 compute squared weights, total and normalised probabilities. Check the sum.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Compare a fixed draw

> Do not infer factual improvement from a changed token.

Your evidence record

- Main table order red,blue,green with probabilities0.6,0.3,0.1. Intervals[0,0.6),[0.6,0.9),[0.9,1). T=0.5 squares and normalises:18/23,9/46,1/46; boundaries18/23,45/46,1. Separate pictured A/B probabilities0.8/0.2 become16/17,1/17. Do not substitute those boundaries into the main table.

- Trace u=0.7 at T=1 and T=0.5. Does the different **output** demonstrate improved factual accuracy?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Specify a controlled experiment

> Fix task, model conditions and evaluation criteria before comparison.

Your evidence record

- Main table order red,blue,green with probabilities0.6,0.3,0.1. Intervals[0,0.6),[0.6,0.9),[0.9,1). T=0.5 squares and normalises:18/23,9/46,1/46; boundaries18/23,45/46,1. Separate pictured A/B probabilities0.8/0.2 become16/17,1/17. Do not substitute those boundaries into the main table.

- A student changes the prompt and model as well as **temperature**, then attributes a better answer to **temperature**. Propose a fair comparison and reject any unsupported universal claim.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Sample a new table independently

> Keep the first attempt before hints.

Your evidence record

- Use exact fractions for squared weights and cumulative boundaries. Intervals are left-inclusive,right-exclusive in A,B,C order. Zero temperature is outside the stated formula. The task provides no factual reference for these labels, so explain the procedural comparison without calling either output true.

- Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Sample a new table independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Compute the adjusted table for **Sampling** and **temperature**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for **Sampling** and **temperature**. Proposed explanation: "**Sampling** means equal chances". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

One prompt produces three different valid quiz questions.

Does **variability** mean failure?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

My reasoning and evidence

▼ Extension · A small attention calculation — W1 Compute scores and weights

REFERENCE · SECTION 17

Extension · A small attention calculation — W1 Compute scores and weights

> Use the declared dimensions, values and score procedure beside this task.

Contract, question and response

- Attention extension reference: q=[1],keys[0],[ln(3)],values[2],[6],width1. Scores0,ln(3); stable softmax subtracts their maximum before exponentiation. For the separate equal-score example q=[1,1],keys[1,0],[0,1],values[2,0],[0,4],divide scores bysqrt(2). Mask before normalisation; reject incompatible key/query widths. These weights do not verify source truth.

- For q=[1], keys [0],[ln(3)] and d=1, compute **score**s and **softmax** weights.

My reasoning and evidence

▼ **Extension · A small attention calculation — W2 Combine and mask**

REFERENCE · SECTION 18

Extension · A small attention calculation — W2 Combine and mask

> Use the declared dimensions, values and score procedure beside this task.

Contract, question and response

- Attention extension reference: q=[1],keys[0],[ln(3)],values[2],[6],width1. Scores0,ln(3); stable softmax subtracts their maximum before exponentiation. For the separate equal-score example q=[1,1],keys[1,0],[0,1],values[2,0],[0,4],divide scores bysqrt(2). Mask before normalisation; reject incompatible key/query widths. These weights do not verify source truth.

- Use values [2],[6] for the preceding weights. Compute **output** and **output** when the second key is excluded.

My reasoning and evidence

▼ **Extension · A small attention calculation — W3 Trace equal scores**

REFERENCE · SECTION 19

Extension · A small attention calculation — W3 Trace equal scores

> Use the declared dimensions, values and score procedure beside this task.

Contract, question and response

- Attention extension reference: q=[1],keys[0],[ln(3)],values[2],[6],width1. Scores0,ln(3); stable softmax subtracts their maximum before exponentiation. For the separate equal-score example q=[1,1],keys[1,0],[0,1],values[2,0],[0,4],divide scores bysqrt(2). Mask before normalisation; reject incompatible key/query widths. These weights do not verify source truth.

- For q=[1,1], keys [1,0],[0,1], values [2,0],[0,4], compute scaled **score**s, weights and **output**.

My reasoning and evidence

▼ Extension · A small attention calculation — W4 Calculate new attention independently

REFERENCE · SECTION 20

Extension · A small attention calculation — W4 Calculate new attention independently

> Retained extension task with its own teacher key and response.

Extension question and record

- q=[1], keys [0],[ln(2)],[ln(3)], values [0],[3],[8], width 1. Compute **score**s, weights and **output**. Then mask the third item and recompute. Explain whether **output** establishes a factual claim.

> **My reasoning and evidence**

▼ Extension · A small attention calculation — W5 Check stability and shape

REFERENCE · SECTION 21

Extension · A small attention calculation — W5 Check stability and shape

> Use the declared dimensions, values and score procedure beside this task.

Contract, question and response

- Attention extension reference: q=[1],keys[0],[ln(3)],values[2],[6],width1. Scores0,ln(3); stable softmax subtracts their maximum before exponentiation. For the separate equal-score example q=[1,1],keys[1,0],[0,1],values[2,0],[0,4],divide scores bysqrt(2). Mask before normalisation; reject incompatible key/query widths. These weights do not verify source truth.

- **Score**s are 1000 and 1000+ln(3); values 4 and 8. Compute stable shifted weights and **output**. What if one key has width two and the query width one?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional two-token distribution illustrates **variability** in generation.

At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

AI ACADEMY / AI-3/4

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.05882352941176471

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/4

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Original distribution

A boundary 0.8; u=0.85 selects B.

↓

2. T=0.5 weights

Square probabilities:0.64 and 0.04; total 0.68.

↓

3. Normalise

A=16/17≈0.9412; B=1/17≈0.0588.

↓

4. Same draw

0.85 is below 16/17, so T=0.5 selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional two-token distribution illustrates **variability** in generation.
- Which labelled step matches this detail? 0.85 is below 16/17, so T=0.5 selects A.
- 1. Normalise
- 2. Original distribution
- 3. T=0.5 weights
- 4. Same draw
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after trying**

After your attempt, compare with the supported explanation: Same draw 0.85 is below 16/17, so T=0.5 selects A. Lower positive **temperature** concentrates this unequal distribution on its larger entry.

It changes **sampling** probabilities, not **evidence** for whether A or B is true. Using the same draw isolates how the distribution changes the decision.

This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Changed-case question

- Changed-case practice: Set the draw to 0.97, keeping both distributions. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Both **temperature**s select B.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

<details>
<summary>Check changed case</summary>

After your attempt, compare with the supported explanation: Both **temperature**s select B. 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.
</details>

Repair and keep evidence

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

> Choose the relevant picture rather than repeating every task.

01. Sampling and Variability: the case

AI ACADEMY / AI-3/4 · 01 · LOOK AT THE FACTS

Sampling and Variability: the case

The situation

A fictional two-token distribution illustrates variability in generation.

Supplied evidence

At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.
Use token order A,B and cumulative intervals [0,boundary),[boundary,1). Draw u=0.85.

By the end, I can calculate a toy temperature change and trace a sampling draw.

Follow the labelled picture with the matching explanation. Open the vector diagram

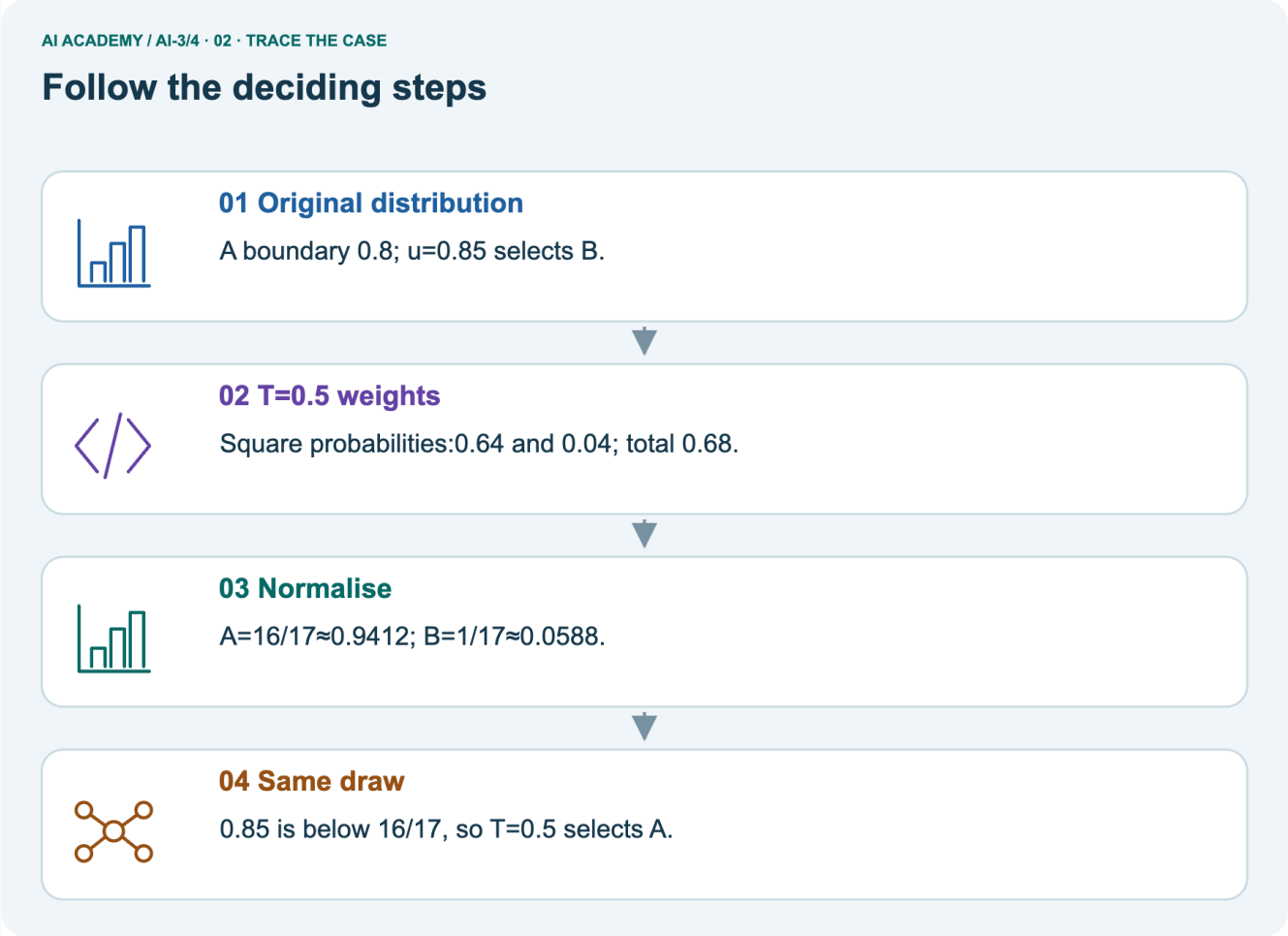
A fictional two-token distribution illustrates **variability** in generation. Supplied facts: At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



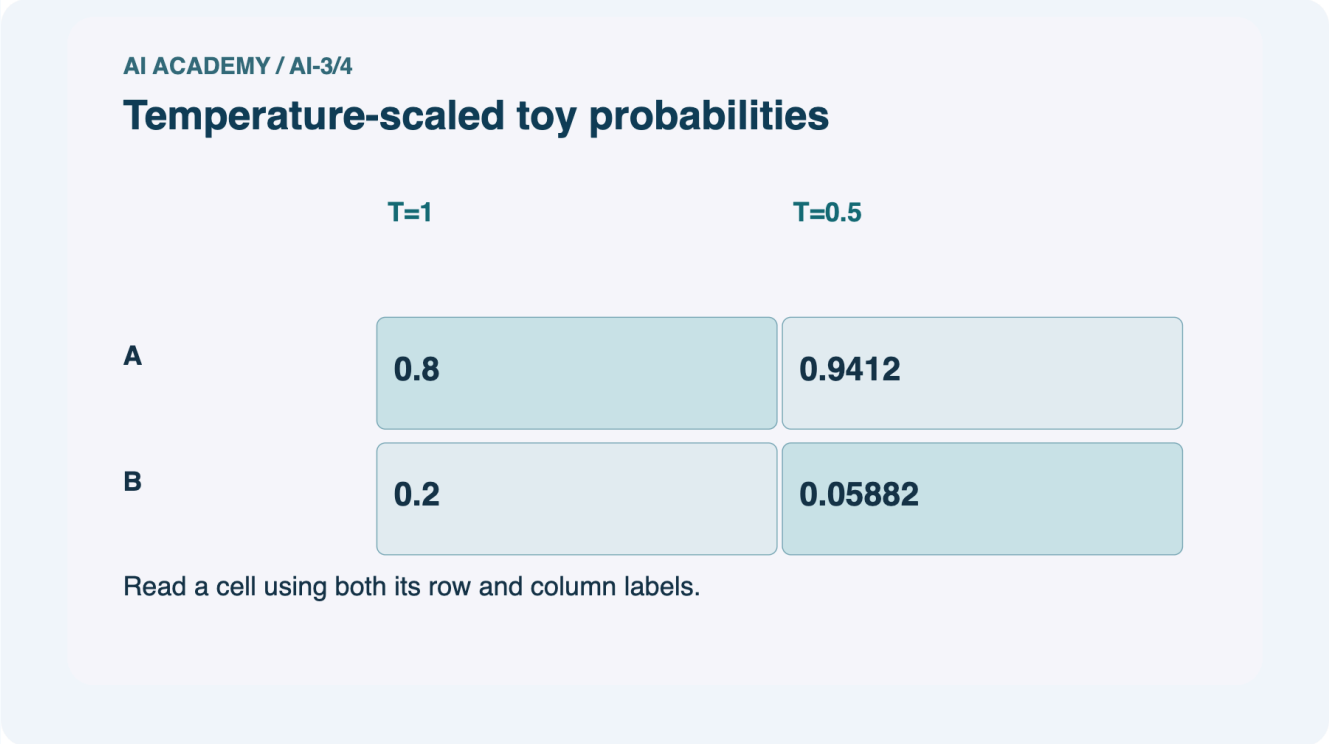
Follow the labelled picture with the matching explanation. Open the vector diagram

Original distribution: A boundary 0.8; u=0.85 selects B. → T=0.5 weights: Square probabilities:0.64 and 0.04; total 0.68. → Normalise: A=16/17≈0.9412; B=1/17≈0.0588. → Same draw: 0.85 is below 16/17, so T=0.5 selects A.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Temperature-scaled toy probabilities



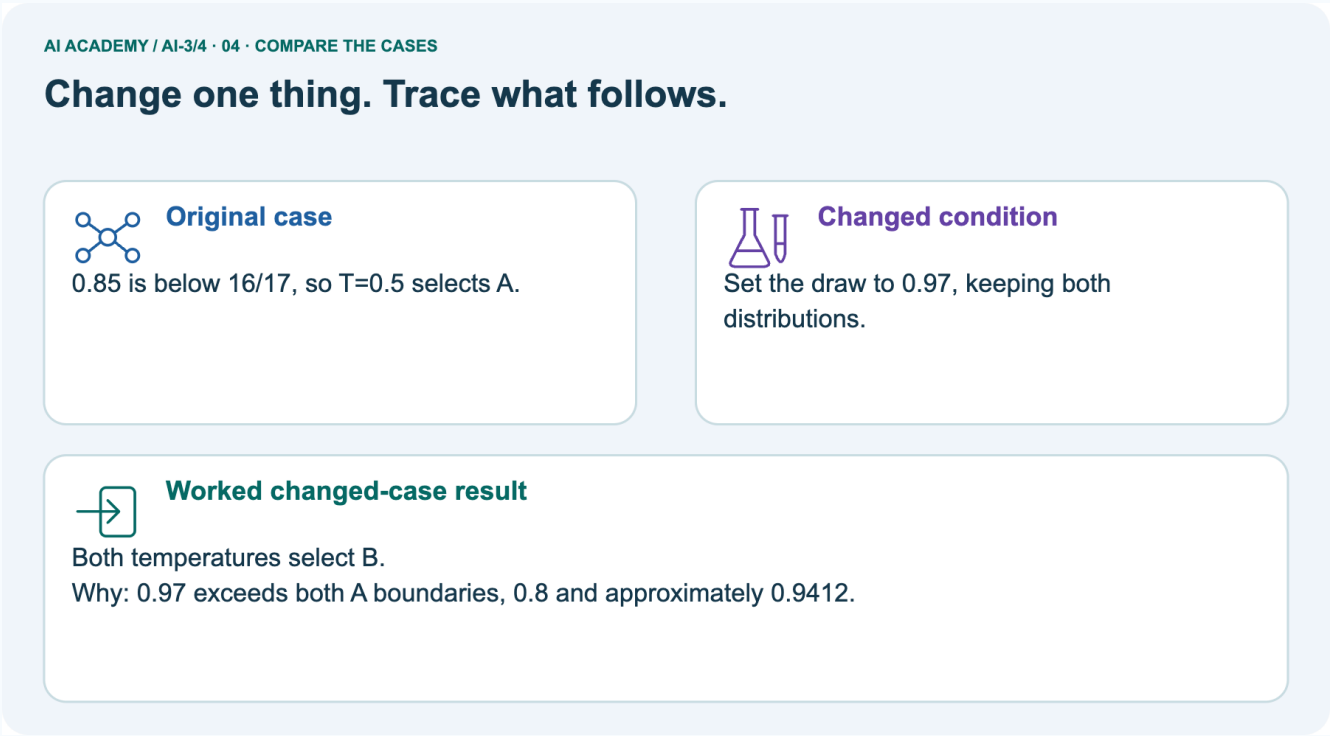
Follow the labelled picture with the matching explanation. Open the vector diagram

Temperature-scaled toy probabilities. A: T=1 = 0.8, T=0.5 = 0.9411764705882353. B: T=1 = 0.2, T=0.5 = 0.058823529411764705.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

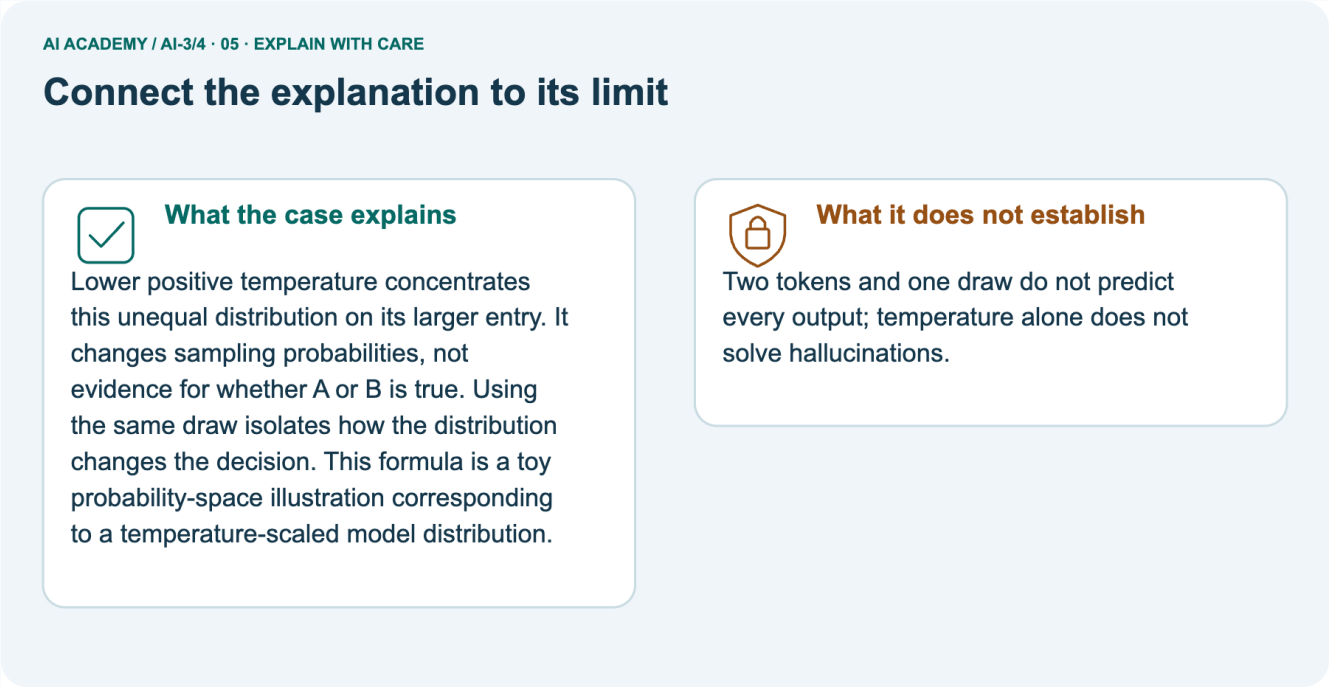
Original case: 0.85 is below 16/17, so T=0.5 selects A. Changed condition: Set the draw to 0.97, keeping both distributions. Expected result: Both **temperature**s select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

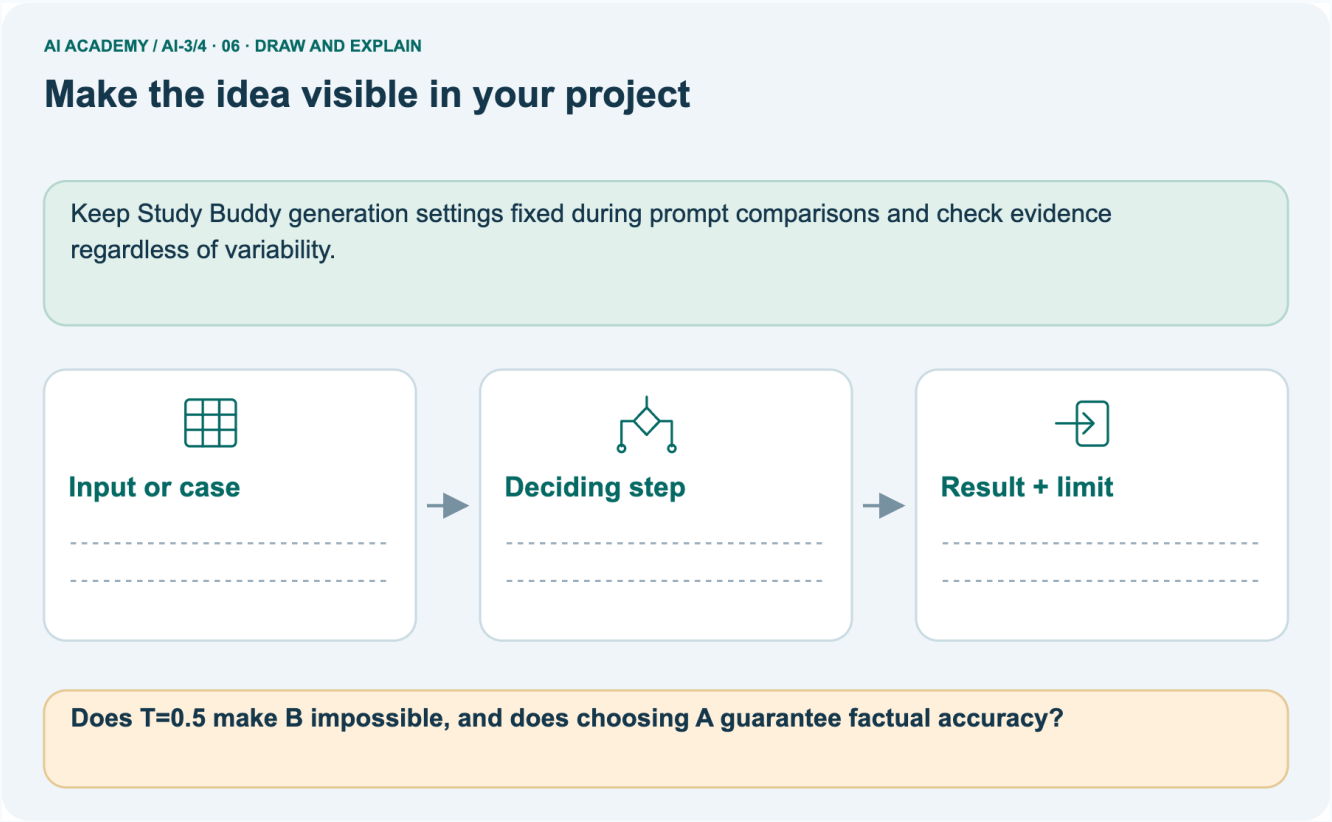
Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Limit: Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**. Draw your **input**, deciding step, result and limit.

Supported question: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sampling and Variability: the case

02. Follow the deciding steps

03. Temperature-scaled toy probabilities

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.